

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO3

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 3

Unit Title: Intelligent Agents

Unit Mapped RBT: L2 (Understand)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Explain what makes an agent rational, and describe the things on which rationality depends. (10 Marks)

- (a) State what a rational agent is required to do, and name the four things that its rationality is judged against.
- (b) Explain why an action can be rational and still turn out badly, using an example.

[L2 - Understand]

Q2. Distinguish rationality from omniscience, and explain what follows from that distinction for the design of an agent. (10 Marks)

- (a) Describe what each of the two would require of an agent.
- (b) Explain why an agent should gather information and explore, rather than act only on what it already believes.

[L2 - Understand]

Q3. Distinguish a simple reflex agent from a model-based reflex agent. (10 Marks)

- (a) Describe how each of the two decides on its next action, and what internal information each one keeps.
- (b) Explain a task on which the simple reflex agent is inadequate, and state exactly what defeats it.

[L2 - Understand]

Q4. Explain why a learning agent must sometimes take actions it does not believe to be best. (10 Marks)

- (a) Describe the component responsible for producing such actions and what it contributes.
- (b) Explain what would happen to the agent if it always acted on its current best belief.

[L2 - Understand]

Q5. Explain what observability means for an agent, and what a partially observable environment demands of its design. (10 Marks)

- (a) Describe the difference between a fully observable and a partially observable environment.
- (b) Explain what the agent must maintain in the partially observable case, and why.

[L2 - Understand]

Q6. Distinguish a goal-based agent from a utility-based agent. (10 Marks)

- (a) Describe what each of the two holds and how each decides between two available actions.
- (b) Explain a situation in which the goal-based agent cannot choose while the utility-based agent can.

[L2 - Understand]

Q7. Describe a multi-agent system and explain what cooperation and coordination each require within one. (10 Marks)

- (a) Describe what makes an environment multi-agent, and distinguish a cooperative setting from a competitive one.
- (b) Explain what can go wrong when several agents each act sensibly on their own, giving one example.

[L2 - Understand]

Q8. Describe two applications of intelligent agents and state the agent type each one calls for. (10 Marks)

- (a) Describe the task in each of the two cases, naming what the agent perceives and what it does.
- (b) Explain which agent architecture each application requires, and give the reason.

[L2 - Understand]

Q9. Describe human-AI collaboration and explain when it is to be preferred to full automation. (10 Marks)

- (a) Describe two patterns by which a person and an intelligent system can share a task.
- (b) Explain the circumstances under which the person must remain in the loop, and why.

[L2 - Understand]

Q10. Explain how the performance measure of an agent should be written, and what goes wrong when it is written badly. (10 Marks)

- (a) State what the performance measure should be expressed in terms of, and why.
- (b) Explain, with an example, how a badly written measure can be satisfied without the task being done.

[L2 - Understand]

----- End of Sample Question Set -----